

- 9V battery
- 15Ω , 150Ω , 470Ω , $1.5k\Omega$
- LED
- DMM + leads

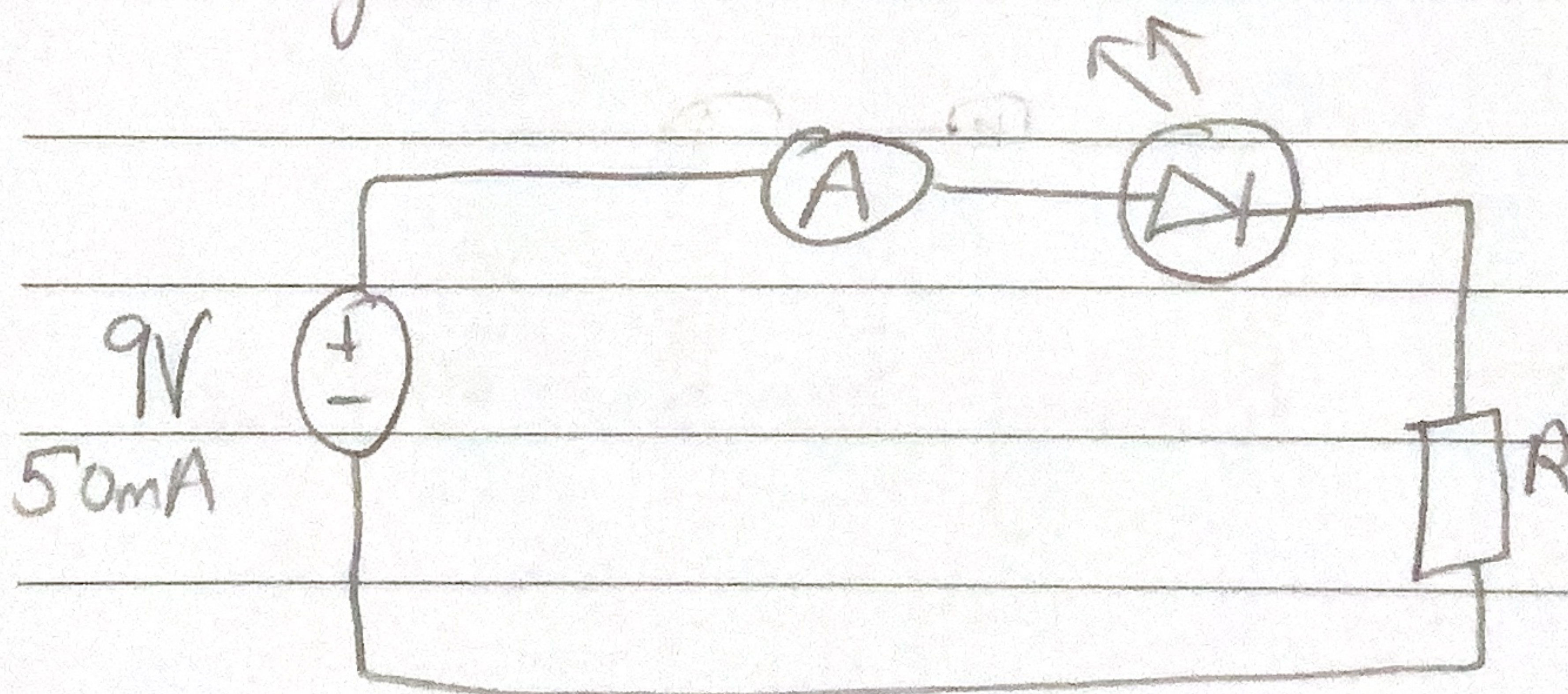
$15\Omega \Rightarrow \text{Reading: } 14.92\Omega$

$150\Omega \Rightarrow \text{Reading: } 147.45$

$470\Omega \Rightarrow \text{Reading: } 466.03\Omega$

$1.5k\Omega \Rightarrow \text{Reading: } 1.5003k\Omega$

Figure 2-11 $\Rightarrow$



When $15k\Omega = R \therefore I = 4.707 \text{ mA}$

When $470\Omega = R \therefore I = 14.914$

$$P = VI \quad V = 9, R = 1.5k\Omega, I \Rightarrow$$

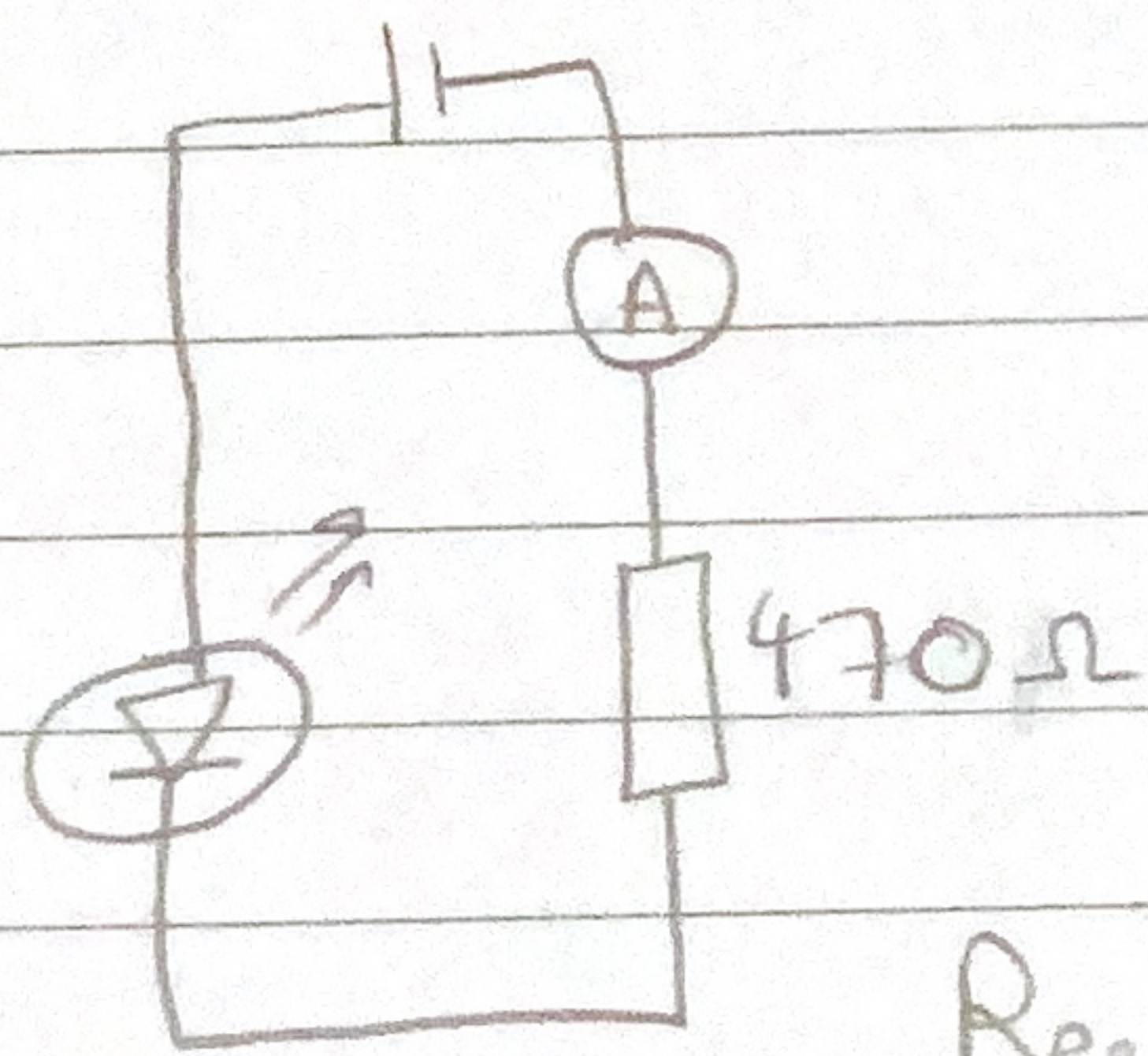
$$P = V^2/R \quad P \Rightarrow$$

$$P = 9^2/1500 = 0.054 = 54 \text{ mW}$$

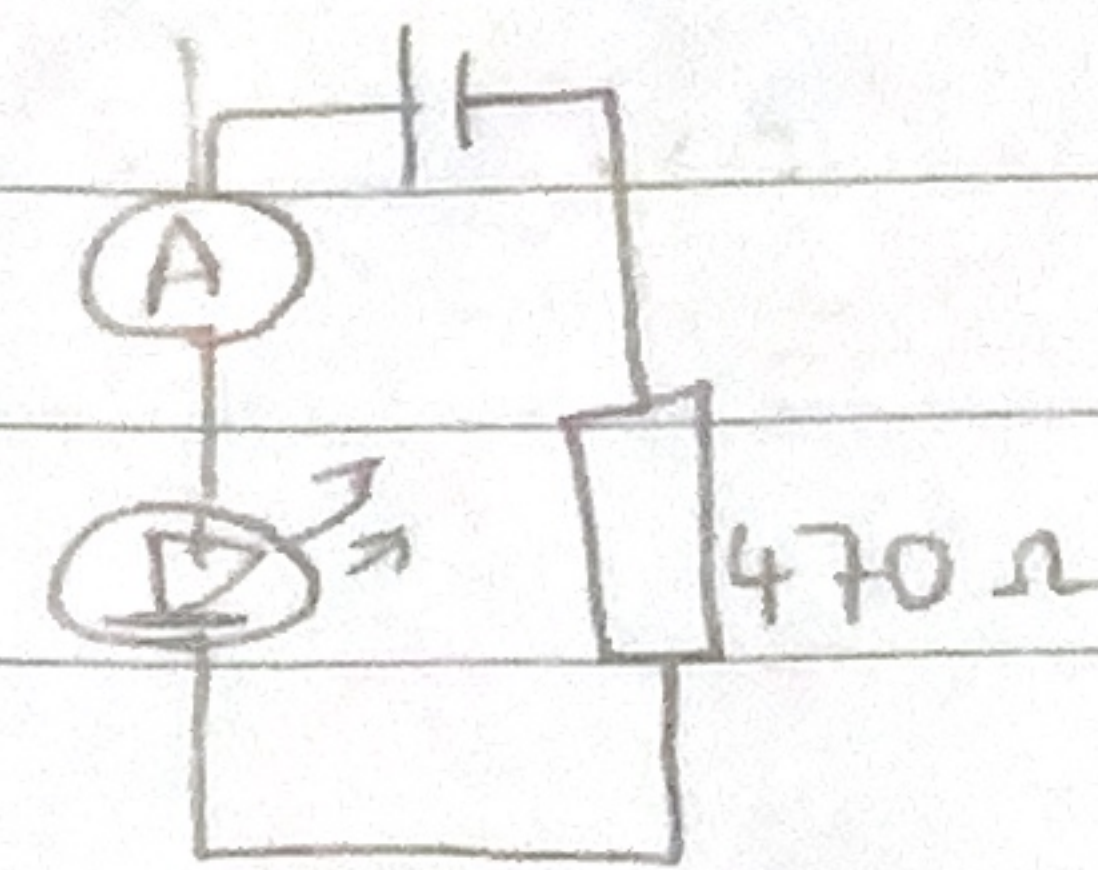
$$I = P/V \Rightarrow 0.006 = 6 \text{ mA}$$

$$\text{Actual } P = I^2 R = 1500 \times (4.707 \times 10^{-3})^2$$

$$= 0.033 \approx 33 \text{ mW}$$



Reading from dmm $\Rightarrow 15 \text{ mA}$



Reading of DMM $\Rightarrow$
15 mA

Reading Voltage across battery $\Rightarrow 8.995 \text{ V}$

$V = 7.056$ across resistor

$V = 1.93$ across LED

$$1.93 + 7.056 \approx 9 \text{ V} \approx 8.995 \text{ V}$$

V_R	$I \text{ (mA)}$	R
6.927	0.21 mA	330 Ω
6.967	0.14 mA	470 Ω
7.03	7.03 mA	1k Ω

Date 11.6.26

Connecting 2 probes to measure DC
voltage